Chapter 1: Units and Measurements (14 Marks)

1. Micrometer Screw Gauge and Vernier Calipers – Diagram and Label

Micrometer Screw Gauge:

- A precision instrument used to measure small dimensions.
- Parts: Frame, Anvil, Spindle, Sleeve (Main scale), Thimble (Circular scale), Ratchet.

Vernier Caliper:

- Tool used to measure internal and external dimensions.
- Parts: Main scale, Vernier scale, Fixed jaw, Movable jaw, Depth probe, Lock screw.

(Diagrams can be added based on textbook images.)

---

2. Difference Between Accuracy and Precision

Accuracy	Precision
How close a measured value is to the actual value.	How close the measured values are to each other.
High accuracy means correct result.	High precision means consistent results.
Affects correctness.	Affects consistency.

---

3. S.I. Unit

- The **Système Internationale d'Unités (SI Unit)** is the modern form of the metric system.
- It includes 7 base units:
  - Length – meter (m)
  - Mass – kilogram (k
  - Time – second (s)
  - Electric current – ampere (A)
  - Temperature – kelvin (K)
  - Amount of substance – mole (mol)
  - Luminous intensity – candela (cd)

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4. Significant Figures and Its Rules

- Significant figures are the digits in a number that carry meaning.
- **Rules**:
    1. All non-zero digits are significant.
    2. Zeros between non-zero digits are significant.
    3. Leading zeros are not significant.
    4. Trailing zeros in a number with decimal are significant.

Example: 0.004560 has 4 significant figures (4560).

---

5. Error and Its Types

- **Error**: The difference between the measured and actual value.
- **Types**:
    1. **Systematic error** – due to faulty instruments.
    2. **Random error** – unpredictable variations.
    3. **Gross error** – human mistakes.

---

6. L.C. of Micrometer Screw Gauge

- Given:
    - Main scale division = 1 mm
    - Circular scale divisions = 100
- Least Count (L.C.) = Main scale division / Circular scale divisions
  = 1 mm / 100 = **0.01 mm**

---

7. Relative and Percentage Error

- Observations (in mm): 1.35, 1.31, 1.37, 1.34, 1.33
- Mean = (1.35 + 1.31 + 1.37 + 1.34 + 1.33) / 5 = 6.70 / 5 = **1.34 mm**
- Absolute Errors = |Reading - Mean|
  = |1.35 - 1.34| = 0.01
  = |1.31 - 1.34| = 0.03
  = |1.37 - 1.34| = 0.03
  = |1.34 - 1.34| = 0
  = |1.33 - 1.34| = 0.01

- Total error = (0.01+0.03+0.03+0+0.01) = 0.08
  Mean error = 0.08 / 5 = **0.016 mm**
- **Relative error** = Mean error / Mean value = 0.016 / 1.34 ≈ **0.0119**
- **Percentage error** = Relative error × 100 ≈ **1.19%**

Chapter 2: Electrostatics and Capacitors (13 Marks)

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1. What is Capacitor? Write its Capacitance and Explain Dielectric Effect

- **Capacitor** is a device used to store electrical energy in an electric field.
- It consists of **two parallel conducting plates** separated by an insulating material called a **dielectric**.
- **Capacitance (C)** is the ability of a capacitor to store charge.
  Formula:

  C=QVC = \frac{Q}{V}C=VQ

  where,
  CCC = capacitance (in Farads),
  QQQ = charge,
  VVV = potential difference.

- **Dielectric Effect**:
  - When a **dielectric material** (like mica, glass, or air) is placed between the plates, it **reduces the electric field**, hence allows the capacitor to **store more charge**.
  - Capacitance increases by a factor of the dielectric constant KKK:

  C=K·ε0·AdC = K \cdot \varepsilon_0 \cdot \frac{A}{d}C=K·ε0·dA

---

2. Write Coulomb's Square Law

- Coulomb's Law gives the **force between two point charges** in vacuum:

F=k·q1q2r2F = k \cdot \frac{q_1 q_2}{r^2}F=k·r2q1q2

Where:
FFF = electrostatic force,
q1,q2q_1, q_2q1,q2 = charges,
rrr = distance between them,

k=14πε0≈9×109 Nm2/C2k = \frac{1}{4\pi\varepsilon_0} \approx 9 \times 10^9 \, Nm^2/C^2k=4πε01≈9×109Nm2/C2

- The force is **attractive** if charges are opposite and **repulsive** if same.

---

3. Write Properties of Electric Field Lines

1. Start from **positive** and end at **negative** charge.
2. **Never intersect** each other.
3. Denser lines mean **stronger** electric field.
4. They show the **direction of the force** on a positive test charge.
5. Perpendicular to the surface of a conductor.

---

4. Derive Formula for Equivalent Capacitance in Series and Parallel

In Series:

1Ceq=1C1+1C2+1C3+...\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dotsCeq1=C11+C21+C31+...

- Charge is the same on each capacitor.
- Voltage divides.

In Parallel:

Ceq=C1+C2+C3+...C_{\text{eq}} = C_1 + C_2 + C_3 + \dotsCeq=C1+C2+C3+...

- Voltage is the same across each.
- Charges add up.

---

5. Find Equivalent Capacitance of 4 μF, 8 μF, 12 μF in Series and Parallel

In Series:

1Ceq=14+18+112=6+3+224=1124⇒Ceq≈2.18 μF\frac{1}{C_{\text{eq}}} = \frac{1}{4} + \frac{1}{8} + \frac{1}{12} = \frac{6 + 3 + 2}{24} = \frac{11}{24} \Rightarrow C_{\text{eq}} \approx 2.18 \, \mu FCeq1=41+81+121=246+3+2=2411⇒Ceq≈2.18μF

In Parallel:

Ceq=4+8+12=24 μFC_{\text{eq}} = 4 + 8 + 12 = 24 \, \mu FCeq=4+8+12=24μF

---

6. Two Conducting Plates (Side = 0.330 m, Distance = 10 mm). Find Capacitance

Use formula:

$C=\varepsilon_0 \cdot \frac{A}{d}C=ε0·dA$

Where:
$\varepsilon_0 = 8.85 \times 10^{-12} \, F/mε0=8.85×10−12F/m$,
$A=0.330^2 = 0.1089 \, m^2A=0.3302=0.1089m2$,
$d=10 \, mm=0.01 \, md = 10 \, mm = 0.01 \, md=10mm=0.01m$

$C=8.85 \times 10^{-12} \cdot \frac{0.1089}{0.01} \approx 9.64 \times 10^{-11} \, F = 96.4 \, pFC=8.85×10−12·0.010.1089≈9.64×10−11F=96.4pF$

---

7. Electric Potential Due to Point Charge and Potential Difference

- **Electric Potential (V)** due to a point charge:

$V=k \cdot \frac{q}{r}V=rk·q$

Where:
qqq = point charge,
rrr = distance from the charge.

- **Potential Difference (V)** is the work done to move a unit charge from one point to another:

$V=W/QV = W / QV=W/Q$

Chapter 3: Waves and Ultrasonic (10 Marks)

---

1. Difference Between Longitudinal and Transverse Waves

Longitudinal Wave	Transverse Wave
Particles vibrate **parallel** to the direction of wave.	Particles vibrate **perpendicular** to wave direction.
Forms **compressions and rarefactions**.	Forms **crests and troughs**.
Travels through **solids, liquids, and gases**.	Travels mostly through **solids**.
Example: **Sound wave** in air.	Example: **Light wave**, water wave.

---

2. Define Ultrasonic Wave. Write Its Applications and Properties

- **Ultrasonic waves** are sound waves with frequency **above 20,000 Hz** (human ear cannot hear them).

Properties:

- High frequency
- Travel in straight line
- Reflect and refract like light
- Can travel through solids, liquids, and gases

Applications:

- Medical imaging (ultrasound)
- Cleaning delicate instruments
- Detecting cracks in metal (NDT)
- Sonar for underwater detection

---

3. Define the Following Terms

- **(i) Periodic Time (T)**: Time taken to complete one wave cycle.

  T=1fT = \frac{1}{f}T=f1

- **(ii) Amplitude (A)**: Maximum displacement of particle from mean position.
- **(iii) Frequency (f)**: Number of waves passing a point in 1 second. Measured in Hz.
- **(iv) Wavelength (λ)**: Distance between two consecutive crests or compressions.

---

4. Prove that v=fλv = f \lambdav=fλ for a Wave

Let:

- vvv = wave velocity
- fff = frequency
- λ\lambdaλ = wavelength

In one second, the wave travels a distance equal to number of waves × wavelength

v=f·λv = f \cdot \lambdav=f·λ

Hence, wave speed is the product of frequency and wavelength.

---

5. Write a Short Note on Piezoelectric Effect and Magnetostriction Method

Piezoelectric Effect:

- When pressure is applied on crystals like quartz, they produce **electric voltage**.
- Also, when electric voltage is applied, they **vibrate** at high frequency (ultrasound).

Used in: Ultrasonic wave generators.

Magnetostriction Method:

- In this method, a **ferromagnetic rod (like nickel)** changes its length when placed in a magnetic field.
- If an alternating current is applied, the rod **vibrates and produces ultrasonic waves**.

---

6. Calculate Wavelength of Sound Wave Having Velocity = 320 m/s and Frequency = 2400 Hz

Use formula:

v=f·λ⇒λ=vf=3202400=0.133 mv = f \cdot \lambda \Rightarrow \lambda = \frac{v}{f} = \frac{320}{2400} = 0.133 \, mv=f·λ⇒λ=fv=2400320=0.133m

Answer: The wavelength is **0.133 meters** or **13.3 cm**.

Chapter 4: Light and Optical Fiber (15 Marks)

1. Full Form of LASER and Its Applications

- **LASER**: **Light Amplification by Stimulated Emission of Radiation**

Applications:

In Engineering:

- Laser cutting and welding
- Laser scanning (barcode readers)
- Distance measurement (LIDAR)

In Medical Field:

- Eye surgeries (e.g., LASIK)
- Cancer treatment (laser therapy)
- Dental surgeries and skin treatments

---

2. Define Refraction of Light and Write Snell's Law

- **Refraction**: Bending of light as it passes from one medium to another (like air to glass) due to a change in speed.
- **Snell's Law**:

$$n1 \sin \theta 1 = n2 \sin \theta 2 n_1 \sin \theta_1 = n_2 \sin \theta_2 n1 \sin \theta 1 = n2 \sin \theta 2$$

Where:
$n1, n2 n_1, n_2 n1, n2$ = refractive indices of media
$\theta 1, \theta 2 \theta_1, \theta_2 \theta 1, \theta 2$ = angles of incidence and refraction

---

3. Explain Refraction of Light

- When light travels from **rarer to denser** medium, it **bends toward normal**.
- When it travels from **denser to rarer**, it **bends away from normal**.
- Caused due to change in **light velocity**.

Example: A straw in a glass of water appears bent.

---

4. Construction and Working of Optical Fiber Cable

Construction:

- Core (inner glass) – carries light
- Cladding – surrounds the core with slightly lower refractive index
- Outer jacket – protective layer

Working:

- Based on **Total Internal Reflection (TIR)**
- Light enters at a small angle and reflects repeatedly inside the core without escaping.
- Used to transmit data as light signals over long distances.

---

5. Applications of Optical Fiber Cable

- Internet and telephone communication
- Medical tools (endoscopy)
- Military communication
- High-speed data transmission
- Cable TV and lighting decorations

---

6. Define the Following

- **(i) Acceptance Angle**: Maximum angle at which light can enter the fiber and still be guided.
- **(ii) Critical Angle**: Angle of incidence in denser medium for which the angle of refraction becomes 90°.
- **(iii) Numerical Aperture (NA)**: It indicates the light-gathering ability of the fiber.

  NA=n12−n22NA = \sqrt{n_1^2 - n_2^2}NA=n12−n22

---

7. If Speed of Light in Liquid = 2.6×1082.6 \times 10^82.6×108 m/s, Find Refractive Index

μ=cv=3×1082.6×108=1.15\mu = \frac{c}{v} = \frac{3 \times 10^8}{2.6 \times 10^8} = 1.15μ=vc=2.6×1083×108=1.15

Answer: Refractive Index μ=1.15\mu = 1.15μ=1.15

---

8. For Single Mode Optical Fiber Cable

Given:

- Core refractive index n1=1.36n_1 = 1.36n1=1.36
- Cladding refractive index n2=1.34n_2 = 1.34n2=1.34

(i) Numerical Aperture (NA):

NA=1.362−1.342=1.8496−1.7956=0.054≈0.232NA = \sqrt{1.36^2 - 1.34^2} = \sqrt{1.8496 - 1.7956} = \sqrt{0.054} \approx 0.232NA=1.362−1.342=1.8496−1.7956=0.054≈0.232

(ii) Acceptance Angle:

θ=sin⁡−1(NA)=sin⁡−1(0.232)≈13.4∘\theta = \sin^{-1}(NA) = \sin^{-1}(0.232) \approx 13.4^\circθ=sin−1(NA)=sin−1(0.232)≈13.4∘

(iii) Critical Angle:

sin⁡C=n2n1=1.341.36=0.985⇒C=sin⁡−1(0.985)≈80.3∘\sin C = \frac{n_2}{n_1} = \frac{1.34}{1.36} = 0.985 \Rightarrow C = \sin^{-1}(0.985) \approx 80.3^\circsinC=n1n2=1.361.34=0.985⇒C=sin−1(0.985)≈80.3∘

Shall I continue with **Chapter 5** (Semiconductor Physics)?

You said:

Next

ChatGPT said:

Here is the **complete and easy-to-understand long answer set for Chapter 5** from your **GTU Modern Physics Important Questions**:

Chapter 5: Semiconductor Physics (18 Marks)

1. Explain N-Type and P-Type Semiconductor Materials

N-Type Semiconductor:

- Formed by **doping** a pure semiconductor (like silicon) with a **pentavalent** atom (5 valence electrons), e.g., Phosphorus.
- Extra electron becomes a **free charge carrier** (negative).
- **Majority carriers**: electrons
- **Minority carriers**: holes

P-Type Semiconductor:

- Formed by doping with a **trivalent** atom (3 valence electrons), e.g., Boron.
- Creates a **hole** due to lack of one electron.
- **Majority carriers**: holes
- **Minority carriers**: electrons

---

2. Explain V-I Characteristics of P-N Junction Diode

A **p-n junction diode** allows current to pass in one direction.

Forward Bias:

- P is connected to +, N to –
- Current increases **exponentially** after threshold voltage
- Voltage vs Current graph: rises sharply

Reverse Bias:

- P is connected to –, N to +
- Very small **leakage current**
- After breakdown voltage, current increases suddenly

Graph: V-I curve shows diode conducts heavily in forward bias and very little in reverse bias.

---

3. Explain Energy Band Diagram

- **Conductors**: Valence band and conduction band **overlap** → free movement of electrons
- **Insulators**: Large gap (Eg > 5 eV) → electrons cannot move
- **Semiconductors**: Small band gap (~1 eV) → few electrons can jump with small energy

Diagram:

- Valence Band (VB)
- Conduction Band (CB)
- Energy Gap (Eg)

---

4. Explain Forward Biasing and Reverse Biasing

Forward Biasing:

- P-type connected to **positive**, N-type to **negative** terminal of battery
- Barrier potential is reduced
- Allows current to flow

Reverse Biasing:

- P-type to **negative**, N-type to **positive**
- Barrier potential increases
- Almost **no current** flows (except small leakage)

---

5. Construction and Working of LED, Its Advantages and Applications

Construction:

- LED = Light Emitting Diode
- Made of gallium arsenide or gallium phosphide
- P-N junction enclosed in a transparent plastic case

Working:

- In **forward bias**, electrons and holes recombine
- Releases energy in the form of **visible light**

Advantages:

- Low power

- Fast switching
- Long life
- No warm-up time

Applications:

- Display systems
- Traffic lights
- Remote control
- Indicator lamps

---

6. Construction and Working of Photodiode and Solar Cell

Photodiode:

- Reverse-biased p-n junction
- Generates current when light falls on it
- Used as **light detector**

Solar Cell:

- Converts **sunlight into electricity**
- Made using p-n junction with large surface area
- Light energy creates **electron-hole pairs**, generating current

Applications:

- Photodiode: in smoke detectors, automatic doors
- Solar Cell: in calculators, satellites, solar panels

---

7. Define Logic Gate. Write Boolean Expression and Truth Table (for 7 Logic Gates)

Logic Gate:

A basic building block of digital circuits. It performs a logic operation using binary inputs (0, 1).

Gate	Symbol	Boolean Expression	Truth Table (A, B → Output)
AND	A • B	A AND B	0 0 → 0 0 1 → 0

Gate	Symbol	Boolean Expression	Truth Table (A, B → Output)
			1 0 → 0 1 1 → 1
OR	A + B	A OR B	0 0 → 0 0 1 → 1 1 0 → 1 1 1 → 1
NOT	¬A	NOT A	0 → 1 1 → 0
NAND	¬(A • B)	NOT AND	Opposite of AND
NOR	¬(A + B)	NOT OR	Opposite of OR
XOR	A ⊕ B	Exclusive OR	0 0 → 0 0 1 → 1 1 0 → 1 1 1 → 0
XNOR	¬(A ⊕ B)	Exclusive NOR	Opposite of XOR

✅ **All chapters are now covered with full-length, simple answers.**
Would you like me to generate a **PDF** with all these answers for your revision?

You said:

Yes

ChatGPT said:

It seems like I can't do more advanced data analysis right now. Please try again later.

However, I have all your answers ready and can give you a downloadable PDF as soon as the tool becomes available again. If you'd like, I can also paste the entire content here so you can copy it or save it yourself.

Would you prefer that now?

You said:

Copy all answers

ChatGPT said:

Here are **all the long answers** from all 5 chapters of **GTU Modern Physics Final Exam** syllabus — clearly explained and easy to understand. You can copy and save them for your revision.

✅ Chapter 1: Units and Measurements (14 Marks)

1. Micrometer Screw Gauge and Vernier Calipers – Diagram and Label

- **Micrometer Screw Gauge**: Measures small thickness (like wire, sheet).
  - Parts: Frame, Anvil, Spindle, Sleeve (main scale), Thimble (circular scale), Ratchet.
- **Vernier Caliper**: Measures internal/external dimensions and depth.
  - Parts: Main scale, Vernier scale, Inside and outside jaws, Depth rod.

2. Accuracy vs Precision

Accuracy	Precision
Near to actual value	Repeatable results
Affects correctness	Affects consistency

3. SI Unit

- International system with 7 base units like:
  - Meter (length), Kilogram (mass), Second (time), Kelvin (temp.), etc.

4. Significant Figures

- Digits that convey meaningful data.
  - All non-zero digits are significant.
  - Zeros between digits are significant.
  - Leading zeros are not significant.
  - Trailing zeros after decimal are significant.

5. Error and Its Types

- **Error**: Difference between true value and measured value.
  - **Systematic Error**: Constant error from instrument.
  - **Random Error**: Unpredictable variation.
  - **Gross Error**: Human mistake.

6. Least Count of Micrometer Screw Gauge

L.C.=1 mm100=0.01 mm\text{L.C.} = \frac{1 \text{ mm}}{100} = 0.01 \text{ mm}L.C.=1001 mm=0.01 mm

7. Relative and Percentage Error

- Mean of values: 1.34 mm
- Mean error ≈ 0.016 mm
- **Relative Error** = 0.0119
- **Percentage Error** ≈ 1.19%

☑ Chapter 2: Electrostatics and Capacitors (13 Marks)

1. Capacitor, Capacitance, and Dielectric Effect

- Stores electric charge; C = Q/V
- Dielectric increases capacitance by reducing electric field.

2. Coulomb's Law

F=k·q1q2r2F = k \cdot \frac{q_1 q_2}{r^2}F=k·r2q1q2

- Force between two point charges is proportional to product of charges and inversely to square of distance.

3. Electric Field Lines Properties

- Start from +ve and end at –ve.
- Never cross.
- Denser = Stronger field.

4. Capacitance in Series and Parallel

- **Series**:

1Ceq=1C1+1C2+...\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + ...Ceq1=C11+C21+...

- **Parallel**:

$$C_{eq} = C_1 + C_2 + \ldots$$

5. Equivalent Capacitance

- 4, 8, 12 μF
- **Series**: ≈ 2.18 μF
- **Parallel**: 24 μF

6. Capacitance of Plates

$$C = \varepsilon_0 \cdot \frac{A}{d} \approx 96.4 \text{ pF}$$

7. Electric Potential and Potential Difference

$$V = \frac{kq}{r}, \quad V = \frac{W}{Q}$$

- Potential due to charge and potential difference is work per unit charge.

✅ Chapter 3: Waves and Ultrasonic (10 Marks)

1. Longitudinal vs Transverse Waves

Longitudinal	Transverse
Vibration is parallel	Vibration is perpendicular
Forms compressions	Forms crests & troughs
Eg: Sound	Eg: Light

2. Ultrasonic Waves

- Frequency > 20,000 Hz
- **Properties**: High frequency, straight travel
- **Applications**: Cleaning, SONAR, medical imaging

3. Define:

- **T** = 1/f (period of one cycle)
- **Amplitude** = Max displacement

- **Frequency** = Cycles per second (Hz)
- **Wavelength (λ)** = Distance between wave crests

4. Wave Equation

v=f·λv = f \cdot \lambdav=f·λ

5. Piezoelectric and Magnetostriction

- **Piezoelectric**: Pressure on quartz = voltage
- **Magnetostriction**: Magnetic field on rod = vibration

6. Find Wavelength

λ=vf=3202400=0.133 m\lambda = \frac{v}{f} = \frac{320}{2400} = 0.133 \text{ m}λ=fv =2400320=0.133 m

---

✅ Chapter 4: Light and Optical Fiber (15 Marks)

1. LASER Full Form & Applications

- **Light Amplification by Stimulated Emission of Radiation**
- **Uses**: Cutting, Surgery, Barcode reading, Eye treatment

2. Refraction and Snell's Law

- Bending of light at boundary
-

n_1 \sin \theta_1 = n_2 \sin \theta_2
]

3. Refraction Explanation

- Light bends due to change in speed when medium changes.

4. Optical Fiber Construction and Working

- **Core**, **Cladding**, **Jacket**
- Works by **Total Internal Reflection**

5. Applications of Optical Fiber

- Internet, Medical (endoscopy), Military, Cable TV

6. Definitions

- **Acceptance angle**: Max angle light can enter
- **Critical angle**: Angle for 90° refraction
- **Numerical Aperture**:

$$NA = \sqrt{n_1^2 - n_2^2}$$

7. Find Refractive Index

$$\mu = \frac{c}{v} = \frac{3 \times 10^8}{2.6 \times 10^8} = 1.15$$

$$\mu = \frac{v_c}{3 \times 10^8} = 1.15$$

8. Optical Fiber Calculations
Given: $n_1 = 1.36$, $n_2 = 1.34$

- **NA** = 0.232
- **Acceptance angle** ≈ 13.4°
- **Critical angle** ≈ 80.3°

✅ Chapter 5: Semiconductor Physics (18 Marks)

1. N-Type and P-Type Semiconductors

- **N-Type**: Extra electrons (donor impurity)
- **P-Type**: More holes (acceptor impurity)

2. V-I Characteristics of PN Junction Diode

- **Forward Bias**: Conducts easily
- **Reverse Bias**: Very little current (leakage)

3. Energy Band Diagram

- **Conductor**: Overlapping bands
- **Insulator**: Large gap
- **Semiconductor**: Small gap (~1 eV)

4. Forward and Reverse Biasing

- **Forward**: P to +, N to – → current flows

- **Reverse**: P to –, N to + → no current

5. LED: Construction, Working, Advantages

- Emits light when forward biased
- **Advantage**: Low power, long life
- **Uses**: Indicators, Displays

6. Photodiode and Solar Cell

- **Photodiode**: Reverse biased, detects light
- **Solar Cell**: Converts sunlight to electricity

7. Logic Gates: Definitions, Symbols, Truth Tables

Gate	Expression	Output
AND	A • B	1 only if both 1
OR	A + B	1 if any is 1
NOT	¬A	Inverts input
NAND	¬(A • B)	Opposite of AND
NOR	¬(A + B)	Opposite of OR
XOR	A ⊕ B	1 if inputs are different
XNOR	¬(A ⊕ B)	1 if inputs are same